

| Assessment | Test | Domain        | Skill                                                                         | Difficulty                                        |
|------------|------|---------------|-------------------------------------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Hard <div><div></div><div></div><div></div></div> |

Question

$x(kx - 56) = -16$

In the given equation,  $k$  is an integer constant. If the equation has no real solution, what is the least possible value of  $k$ ?

Correct Answer: 50

Rationale

The correct answer is **50**. An equation of the form  $ax^2 + bx + c = 0$ , where  $a$ ,  $b$ , and  $c$  are constants, has no real solutions if and only if its discriminant,  $b^2 - 4ac$ , is negative. Applying the distributive property to the left-hand side of the equation  $x(kx - 56) = -16$  yields  $kx^2 - 56x = -16$ . Adding **16** to each side of this equation yields  $kx^2 - 56x + 16 = 0$ . Substituting  $k$  for  $a$ ,  $-56$  for  $b$ , and **16** for  $c$  in  $b^2 - 4ac$  yields a discriminant of  $(-56)^2 - 4(k)(16)$ , or **3,136**  $- 64k$ . If the given equation has no real solution, it follows that the value of **3,136**  $- 64k$  must be negative. Therefore, **3,136**  $- 64k < 0$ . Adding **64k** to both sides of this inequality yields **3,136**  $< 64k$ . Dividing both sides of this inequality by **64** yields **49**  $< k$ , or  $k > 49$ . Since it's given that  $k$  is an integer, the least possible value of  $k$  is **50**.